

1. **2G and 3G systems can both transfer data. Compare these approaches with DAB/DVB and list reasons for and against the use of DAB/DVB.**

DAB (Digital Audio Broadcasting) and DVB (Digital Video Broadcasting) are advanced broadcasting systems that can send data at much higher speeds than 2G or 3G mobile networks. However, they only work in one direction — from the transmitter to the receiver — meaning users can receive data but not send it back. These systems are excellent for delivering the same information to a large number of users at once, such as radio programs, television broadcasts, software updates, or popular online content. This makes them very efficient for mass communication.

2. **Which web pages would be appropriate for distribution via DAB or DVB?**

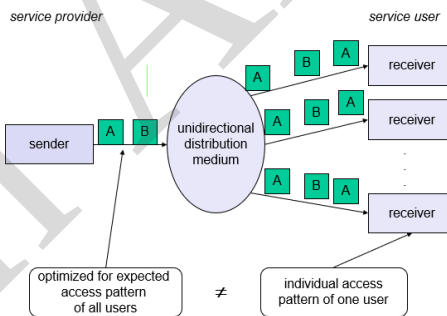
Broadcast systems like DAB and DVB are mainly used to share common information with many users, such as news, weather updates, or popular websites. They can work alongside 2G/3G networks — where the broadcast handles large, common data, and the mobile network handles personal or private data. For example, a news video or advertisement can be sent via broadcast, while the user's personal messages or searches go through the mobile network. Another advantage is that broadcast systems can still work at full speed even when users are moving fast, whereas mobile networks tend to slow down at higher speeds.

3. **How could location based services and broadcast systems work together?**

Location-Based Services (LBS) provide personalized services depending on where the user is located. For example, they can help users find the nearest restaurant, ATM, or bookstore. If the system knows a user's location, it can also combine this with broadcasting — for instance, if several people are standing near a museum, the system could automatically show a video or information about the museum's current exhibition on their devices. This makes services more interactive, relevant, and useful for users based on their surroundings.

❖ **Unidirectional distribution system:**

- ❑ High bandwidth from server to client (downstream), but no bandwidth vice-versa (upstream)
- ❑ Problems of unidirectional broadcast systems
 - A sender can optimize transmitted information only for one group of users/terminals
 - Functions needed to individualize personal requirements/applications



❖ **Define data carousel? Give a short explanation on different types of data carousel.**

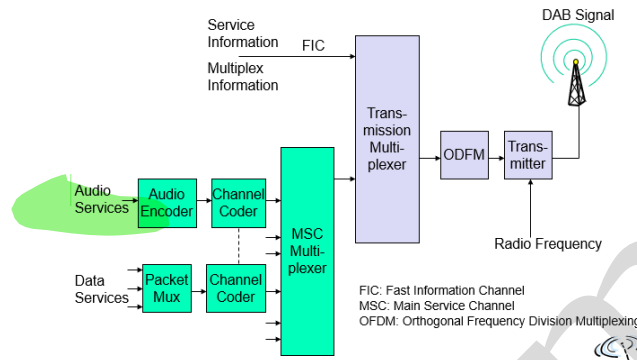
A **data carousel** is a broadcasting technique where the same set of data is repeatedly transmitted in a continuous cycle.

Types of Data Carousel:

1. **Regular (Static) Data Carousel:** The same data set is repeated continuously without any change.
Example: Broadcasting a fixed news bulletin or software update file again and again.
2. **Dynamic Data Carousel:** The content changes over time — new or updated data is added, and old data is removed.
Example: Regularly updated weather reports or live event information.
3. **Adaptive Data Carousel:** The transmission frequency of data items depends on their importance or demand — important data is sent more often.
Example: Frequently accessed web content or popular video clips being broadcast more repeatedly.

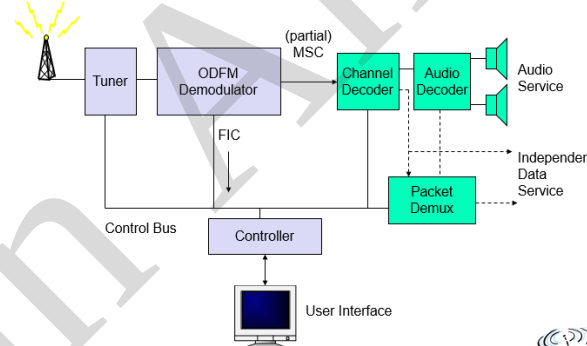
❖ **Illustrate the components of a DAB sender.**

A DAB sender begins with **Audio Services**, which are compressed by an **Audio Encoder**, and **Data Services**, which are organized by a **Packet Mux**. **Service Information** defines the ensemble and is used to build the **FIC (Fast Information Channel)**. The encoded audio and packetized data then pass through their respective **Channel Coders** to add error protection. The protected streams are combined by the **MSC Multiplexer** into the **Main Service Channel (MSC)**. The FIC and MSC are then merged in the **Transmission Multiplexer** to form the complete **DAB Signal**. This signal is processed by the **OFDM (Orthogonal Frequency Division Multiplexing) modulator** to create a robust waveform, which is finally amplified and upconverted by the **Transmitter** to the desired **Radio Frequency** for broadcast.



❖ **Illustrate the components of a DAB receiver.**

A DAB receiver begins by demodulating the incoming signal to separate the **FIC (Fast Information Channel)** and the **MSC (Main Service Channel)**. The **Controller**, managed via the **User Interface**, uses the service list from the FIC to know what's available. When the user selects a service, the controller directs the **MSC data** through a **Channel Decoder** to correct errors. For an **Audio Service**, the data is then sent to an **Audio Decoder** for playback. For an **Independent Data Service**, the data is processed by a **Packet Demux** to extract the specific application. All components are coordinated via the **Control Bus**.



❖ **A problem which "technologies like DAB" are facing is the broad range of different receiver capabilities. Explain a standard solution to solve this problem.**

Problem: DAB broadcasts face a challenge because **different receivers can handle different features** — some can play high-quality audio, others only basic sound.

Solution: Scalable or Layered Coding

- The broadcast is split into **layers**:
 - **Base layer**: basic audio/data that all receivers can use.
 - **Enhancement layers**: extra quality or services for advanced receivers.
- **Result:**
 - All receivers get the main content.
 - Advanced receivers get better quality or extra features.

❖ **What is MOT protocol? Why is it needed? Describe MOT object structure.**

MOT protocol: MOT (Multimedia Object Transfer) protocol is a standard used in DAB (Digital Audio Broadcasting) to transmit multimedia objects, like images, slides, text, or software, alongside audio.

Why Needed: DAB mainly transmits audio, but MOT allows extra content to reach all receivers reliably, even if they join mid-broadcast.

MOT Structures:

MOT formats

- ❑ MHEG, Java, JPEG, ASCII, MPEG, HTML, HTTP, BMP, GIF, ...

Header core

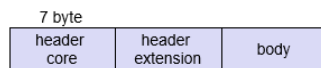
- ❑ size of header and body, content type

Header extension

- ❑ handling information, e.g., repetition distance, segmentation, priority
- ❑ information supports caching mechanisms

Body

- ❑ arbitrary data



- ❑ DAB allows for many repetition schemes
- ❑ objects, segments, headers

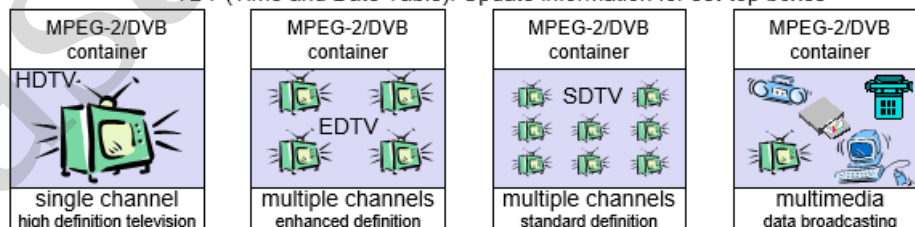
❖ **How the MOT protocol solve the problem faced for broad range of different receiver capabilities?**

- ❑ common standard for data transmission: MOT
- ❑ important for MOT is the support of data formats used in other multimedia systems (e.g., online services, Internet, CD-ROM)
- ❑ DAB can therefore transmit HTML documents from the WWW with very little additional effort.

❖ **DVB Container**

DVB transmits MPEG-2 container

- ❑ high flexibility for the transmission of digital data
- ❑ no restrictions regarding the type of information
- ❑ DVB Service Information specifies the content of a container
 - NIT (Network Information Table): lists the services of a provider, contains additional information for set-top boxes
 - SDT (Service Description Table): list of names and parameters for each service within a MPEG multiplex channel
 - EIT (Event Information Table): status information about the current transmission, additional information for set-top boxes
 - TDT (Time and Date Table): Update information for set-top boxes



❖ **How could location-based services and broadcast systems work together?**

Broadcast systems send data to many users at once, and location-based services (LBS) use the user's location to show relevant info. Together, broadcasts can deliver local updates (like traffic, weather, or alerts) efficiently, and devices pick only what applies to their location.

See the questions three answers